

Rationalizing Denominators: Advanced Mastery

How to add or subtract fractions with surds in their denominators

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Session Flow: Rationalizing Denominators (100 min)

A 100-minute student-centered sequence based on Buckingham's PGC Framework.

Lesson Structure Overview

- **15 min – Phase 1: Novelty & Curiosity** (The LCM Dilemma & Geometric Origins)
- **30 min – Phase 2: Expectation & Guided Discovery** (Square, n -th Roots & Conjugate Proofs)
- **35 min – Phase 3: Effort & Scaffolding** (4-Level Progressive Workshop & Error Analysis)
- **20 min – Phase 4: Recognition & Synthesis** (Peer Assessment Protocol & Exit Ticket)

1. Novelty & Curiosity | The Common Denominator Dilemma

Opening Inquiry Challenge

"Try to combine these two fractions into a single fraction:"

$$\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}}$$

The Wall: Finding the LCM?

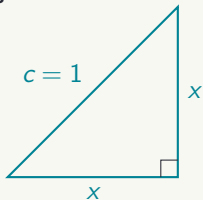
- What is $\text{LCM}(\sqrt{2}, \sqrt{3})$?
- Can we easily find common multiples of surds?
- Maybe we could transform the fractions into a more manageable form.

1. Novelty & Curiosity | Where do these fractions originate?

Real-World Context: Geometry & Trigonometry

Fractions with roots in the denominator occur naturally when solving for sides or ratios:

Right Isosceles Triangle

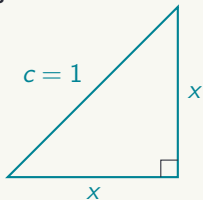


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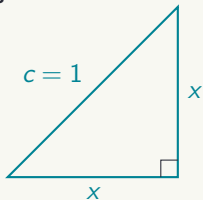
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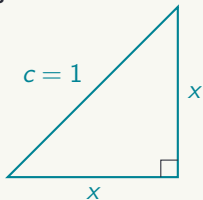
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Core Takeaway

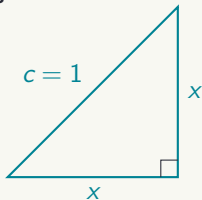
$\frac{1}{\sqrt{2}}$ isn't an artificial trap, it's a natural geometric result.

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Core Takeaway

$\frac{1}{\sqrt{2}}$ isn't an artificial trap, it's a natural geometric result.

Definition: Rationalization

Rationalization is the process of rewriting a fraction so that its denominator contains **only rational numbers** (no radicals), without altering the original value of the expression.

2. Guided Discovery | Target 1: Square Roots in Denominators

Case 1: Radical Index 2 – Denominators of the form $\frac{a}{c\sqrt{b}}$

Fundamental Axiom: Multiplying any term by $1 = \frac{k}{k}$ preserves its value:

2. Guided Discovery | Target 1: Square Roots in Denominators

Case 1: Radical Index 2 – Denominators of the form $\frac{a}{c\sqrt{b}}$

Fundamental Axiom: Multiplying any term by $1 = \frac{k}{k}$ preserves its value:

$$5 = 5 \cdot 1 = 5 \cdot \frac{3}{3} = 5 \cdot \frac{\pi}{\pi} = 5 \cdot \frac{\sqrt{2}}{\sqrt{2}} = 5 \cdot \frac{\sqrt{b}}{\sqrt{b}}$$

3-Step Technique

1. **Identify Form of 1:** Choose $\frac{\sqrt{b}}{\sqrt{b}}$ using the denominator's radical.
2. **Multiply:** Multiply numerators and denominators.
3. **Simplify:** Use $\sqrt{b} \cdot \sqrt{b} = \sqrt{b^2} = (\sqrt{b})^2 = b$ and reduce terms.

General Derivation

$$\frac{a}{c\sqrt{b}} = \frac{a}{c\sqrt{b}} \cdot \left(\frac{\sqrt{b}}{\sqrt{b}} \right) = \frac{a\sqrt{b}}{c \cdot (\sqrt{b})^2} = \frac{a\sqrt{b}}{cb}$$

2. Guided Discovery | Target 1: Worked Examples

[Level 1: Easy]

$$\frac{5}{\sqrt{3}} \xrightarrow{\text{Steps 1 \& 2}} \frac{5}{\sqrt{3}} \cdot \left(\frac{\sqrt{3}}{\sqrt{3}} \right) \xrightarrow{\text{Step 3}} \frac{5\sqrt{3}}{(\sqrt{3})^2} = \frac{5\sqrt{3}}{3}$$

2. Guided Discovery | Target 1: Worked Examples

[Level 1: Easy]

$$\frac{5}{\sqrt{3}} \xrightarrow{\text{Steps 1 \& 2}} \frac{5}{\sqrt{3}} \cdot \left(\frac{\sqrt{3}}{\sqrt{3}} \right) \xrightarrow{\text{Step 3}} \frac{5\sqrt{3}}{(\sqrt{3})^2} = \frac{5\sqrt{3}}{3}$$

[Level 2: Medium]

$$\frac{6}{5\sqrt{3}} \xrightarrow{\text{Steps 1 \& 2}} \frac{6}{5\sqrt{3}} \cdot \left(\frac{\sqrt{3}}{\sqrt{3}} \right) = \frac{6\sqrt{3}}{5 \cdot 3} \xrightarrow{\text{Step 3}} \frac{6\sqrt{3}}{15} = \frac{2\sqrt{3}}{5}$$

2. Guided Discovery | Target 1: Worked Examples

[Level 1: Easy]

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[Level 3: Hard]

$$\frac{4a}{\sqrt{8}} \xrightarrow{\text{Simplify } \sqrt{8}} \frac{4a}{2\sqrt{2}} = \frac{2a}{\sqrt{2}} \xrightarrow{\text{Steps 1, 2 \& 3}} \frac{2a}{\sqrt{2}} \cdot \left(\frac{\sqrt{2}}{\sqrt{2}} \right) = \frac{\cancel{2}a\sqrt{2}}{\cancel{2}} = a\sqrt{2}$$

2. Guided Discovery | Target 2: Monomial Higher-Index Radicals

Case 2: Radical Index n – Denominators of the form $\frac{a}{c\sqrt[n]{b}}$

Core Axiom: To eliminate $\sqrt[n]{b}$, the exponent inside must reach n :

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Core Axiom: To eliminate $\sqrt[n]{b}$, the exponent inside must reach n :

$$\sqrt[n]{b} \cdot \sqrt[n]{b^{n-1}} = \sqrt[n]{b^{1+(n-1)}} = \sqrt[n]{b^n} = b$$

3-Step Technique

1. **Identify Form of 1:** Choose $\frac{\sqrt[n]{b^{n-1}}}{\sqrt[n]{b^{n-1}}}$ using the denominator's radical.
2. **Choose Form of 1:** Multiply top and bottom by $\frac{\sqrt[n]{b^{n-1}}}{\sqrt[n]{b^{n-1}}}$.
3. **Simplify:** Use $\sqrt[n]{b} \cdot \sqrt[n]{b^{n-1}} = \sqrt[n]{b^n} = b$ and reduce terms.

General Derivation

$$\frac{a}{c\sqrt[n]{b}} = \frac{a}{c\sqrt[n]{b}} \cdot \left(\frac{\sqrt[n]{b^{n-1}}}{\sqrt[n]{b^{n-1}}} \right) = \frac{a\sqrt[n]{b^{n-1}}}{c \cdot \sqrt[n]{b^n}} = \frac{a\sqrt[n]{b^{n-1}}}{cb}$$

2. Guided Discovery | Target 2: Worked Examples

[Level 1: Easy]

$$\frac{1}{\sqrt[3]{5}} \xrightarrow{\text{Steps 1 \& 2}} \frac{1}{\sqrt[3]{5}} \cdot \left(\frac{\sqrt[3]{5^2}}{\sqrt[3]{5^2}} \right) \xrightarrow{\text{Step 3}} \frac{\sqrt[3]{25}}{\sqrt[3]{5^3}} = \frac{\sqrt[3]{25}}{5}$$

2. Guided Discovery | Target 2: Worked Examples

[Level 1: Easy]

$$\frac{1}{\sqrt[3]{5}} \xrightarrow{\text{Steps 1 \& 2}} \frac{1}{\sqrt[3]{5}} \cdot \left(\frac{\sqrt[3]{5^2}}{\sqrt[3]{5^2}} \right) \xrightarrow{\text{Step 3}} \frac{\sqrt[3]{25}}{\sqrt[3]{5^3}} = \frac{\sqrt[3]{25}}{5}$$

[Level 2: Medium]

$$\frac{6}{\sqrt[4]{2}} \xrightarrow{\text{Steps 1 \& 2}} \frac{6}{\sqrt[4]{2}} \cdot \left(\frac{\sqrt[4]{2^3}}{\sqrt[4]{2^3}} \right) = \frac{6x\sqrt[4]{2^3}}{2} = 3x\sqrt[4]{8}$$

2. Guided Discovery | Target 2: Worked Examples

[Level 1: Easy]

$$\frac{1}{\sqrt[3]{5}} \xrightarrow{\text{Steps 1 \& 2}} \frac{1}{\sqrt[3]{5}} \cdot \left(\frac{\sqrt[3]{5^2}}{\sqrt[3]{5^2}} \right) \xrightarrow{\text{Step 3}} \frac{\sqrt[3]{25}}{\sqrt[3]{5^3}} = \frac{\sqrt[3]{25}}{5}$$

[Level 2: Medium]

$$\frac{6}{\sqrt[4]{2}} \xrightarrow{\text{Steps 1 \& 2}} \frac{6}{\sqrt[4]{2}} \cdot \left(\frac{\sqrt[4]{2^3}}{\sqrt[4]{2^3}} \right) = \frac{6 \times \sqrt[4]{2^3}}{2} = 3 \times \sqrt[4]{8}$$

[Level 3: Hard]

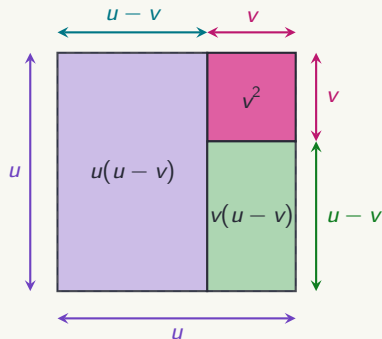
$$\frac{4}{\sqrt[5]{4}} \xrightarrow{\text{Steps 1 \& 2}} \frac{4}{\sqrt[5]{4}} \cdot \left(\frac{\sqrt[5]{4^4}}{\sqrt[5]{4^4}} \right) = \frac{4 \sqrt[5]{4^4}}{\sqrt[5]{4^5}} \xrightarrow{\text{Step 3}} \frac{4 \sqrt[5]{4^4}}{4} = \sqrt[5]{4^4} \xrightarrow{\text{Simplify}} \sqrt[5]{(2^2)^4} = \sqrt[5]{2^8} = 2 \sqrt[5]{2^3} = 2 \sqrt[5]{8}$$

2. Guided Discovery | Target 3: Geometric Proof of Conjugates

Algebraic Identity: $(u - v)(u + v) = u^2 - v^2$

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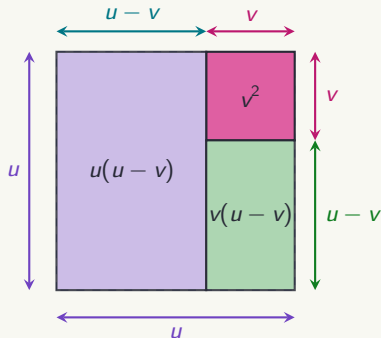
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Area remaining = $(u - v)(u + v)$

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Area remaining = $(u - v)(u + v)$

Geometric Proof Insight

The area of the large square is u^2

$$u^2 - v^2 = u(u - v) + v(u - v) = (u - v)(u + v)$$

Why Conjugates Work

If $u = \sqrt{x}$ and $v = \sqrt{y}$:

$$(\sqrt{x} - \sqrt{y})(\sqrt{x} + \sqrt{y}) = (\sqrt{x})^2 - (\sqrt{y})^2 = x - y$$

2. Guided Discovery | Target 3: Binomial Denominators & Conjugates

Case 3: Binomial Denominators of the form $\frac{a}{\sqrt{b} \pm \sqrt{c}}$ or $\frac{a}{b \pm \sqrt{c}}$

Fundamental Axiom: Difference of squares eliminates cross-terms:

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$$(\sqrt{u} - \sqrt{v})(\sqrt{u} + \sqrt{v}) = (\sqrt{u})^2 - (\sqrt{v})^2 = u - v$$

3-Step Technique

1. **Identify Conjugate:** Flip middle sign ($\sqrt{b} \pm \sqrt{c} \rightarrow \sqrt{b} \mp \sqrt{c}$).
2. **Choose Form of 1:** Multiply top and bottom by $\frac{\text{Conjugate}}{\text{Conjugate}}$.
3. **Simplify Denominator:** Apply $(u - v)(u + v) = u^2 - v^2$ and reduce.

General Derivation: $\sqrt{b} + \sqrt{c}$

$$\begin{aligned}\frac{a}{\sqrt{b} + \sqrt{c}} &= \frac{a}{\sqrt{b} + \sqrt{c}} \cdot \left(\frac{\sqrt{b} - \sqrt{c}}{\sqrt{b} - \sqrt{c}} \right) \\ &= \frac{a(\sqrt{b} - \sqrt{c})}{(\sqrt{b})^2 - (\sqrt{c})^2} \\ &= \frac{a(\sqrt{b} - \sqrt{c})}{b - c}\end{aligned}$$

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- 3. Simplify Denominator:** Apply
($u - v$)($u + v$) = $u^2 - v^2$ and reduce.

General Derivation: $\sqrt{b} - \sqrt{c}$

$$\begin{aligned}\frac{a}{\sqrt{b} - \sqrt{c}} &= \frac{a}{\sqrt{b} - \sqrt{c}} \cdot \left(\frac{\sqrt{b} + \sqrt{c}}{\sqrt{b} + \sqrt{c}} \right) \\ &= \frac{a(\sqrt{b} + \sqrt{c})}{(\sqrt{b})^2 - (\sqrt{c})^2} \\ &= \frac{a(\sqrt{b} + \sqrt{c})}{b - c}\end{aligned}$$

2. Guided Discovery | Target 3: Worked Examples

[Level 1: Easy]

$$\frac{5}{\sqrt{7} - \sqrt{2}} \xrightarrow{\text{Steps 1 \& 2}} \frac{5}{\sqrt{7} - \sqrt{2}} \cdot \left(\frac{\sqrt{7} + \sqrt{2}}{\sqrt{7} + \sqrt{2}} \right) \xrightarrow{\text{Step 3}} \frac{5(\sqrt{7} + \sqrt{2})}{7 - 2} = \frac{\cancel{5}(\sqrt{7} + \sqrt{2})}{\cancel{5}} = \sqrt{7} + \sqrt{2}$$

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[Level 2: Medium]

$$\frac{3}{2 + \sqrt{3}} \xrightarrow{\text{Steps 1 \& 2}} \frac{3}{2 + \sqrt{3}} \cdot \left(\frac{2 - \sqrt{3}}{2 - \sqrt{3}} \right) \xrightarrow{\text{Step 3}} \frac{3(2 - \sqrt{3})}{2^2 - (\sqrt{3})^2} = \frac{6 - 3\sqrt{3}}{4 - 3} = 6 - 3\sqrt{3}$$

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$$\frac{3}{2 + \sqrt{3}} \xrightarrow{\text{Steps 1 \& 2}} \frac{3}{2 + \sqrt{3}} \cdot \left(\frac{2 - \sqrt{3}}{2 - \sqrt{3}} \right) \xrightarrow{\text{Step 3}} \frac{3(2 - \sqrt{3})}{2^2 - (\sqrt{3})^2} = \frac{6 - 3\sqrt{3}}{4 - 3} = 6 - 3\sqrt{3}$$

[Level 3: Hard]

$$\frac{\sqrt{6}}{\sqrt{3} - \sqrt{2}} \xrightarrow{\text{Steps 1 \& 2}} \frac{\sqrt{6}}{\sqrt{3} - \sqrt{2}} \cdot \left(\frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} + \sqrt{2}} \right) = \frac{\sqrt{18} + \sqrt{12}}{3 - 2} \xrightarrow{\text{Simplify Radicals}} 3\sqrt{2} + 2\sqrt{3}$$

3. Effort & Scaffolding | Level 1: Foundation (8 min)

[Level 1: Basic – Individual Work]

Goal: Master basic monomial rationalization and conjugate selection.

Task A: Monomial Terms

1. Rationalize: $\frac{7}{\sqrt{5}}$

2. Rationalize: $\frac{9}{3\sqrt{2}}$

Task B: Identify Conjugate Pair

1. $\sqrt{11} + 3 \rightarrow$

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Task B: Identify Conjugate Pair

1. $\sqrt{11} + 3 \rightarrow \sqrt{11} - 3$
2. $4\sqrt{2} - \sqrt{5} \rightarrow$

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1. $\sqrt{11} + 3 \rightarrow \sqrt{11} - 3$
2. $4\sqrt{2} - \sqrt{5} \rightarrow 4\sqrt{2} + \sqrt{5}$
3. $1 - 3\sqrt{x} \rightarrow$

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3. $1 - 3\sqrt{x} \rightarrow 1 + 3\sqrt{x}$

Solution Check: Task A

$$\frac{7\sqrt{5}}{5} \quad | \quad \frac{3\sqrt{2}}{2}$$

3. Effort & Scaffolding | Level 2: Intermediate & Error Analysis (9 min)

[Level 2: Medium – Pair Work]

Goal: Process higher roots, binomial denominator distribution, and locate algebraic traps.

Problem Set:

1. Rationalize: $\frac{10}{\sqrt[4]{2^3}}$

2. Rationalize: $\frac{\sqrt{3}}{\sqrt{5} + \sqrt{3}}$

Spot the Error!

A student submitted:

$$\frac{6}{\sqrt{x} + 2} = \frac{6\sqrt{x} - 12}{x - 2}$$

Question: Where did the student fail in applying the difference of squares?

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A student submitted:

$$\frac{6}{\sqrt{x} + 2} = \frac{6\sqrt{x} - 12}{x - 2}$$

Question: Where did the student fail in applying the difference of squares?

Correction

Denominator must be $(\sqrt{x})^2 - (2)^2 = x - 4$. Correct answer: $\frac{6\sqrt{x}-12}{x-4}$.

3. Effort & Scaffolding | Level 3: Advanced Binomials (9 min)

[Level 3: Hard – Cooperative Groups]

Goal: Simplify combined fractions and algebraic variables.

Task 1: Addition of Rational Fractions

Simplify completely:

$$\frac{2}{\sqrt{5} - \sqrt{3}} + \frac{2}{\sqrt{5} + \sqrt{3}}$$

Task 2: Algebraic Cancellation

Simplify fully for $a \neq b, a, b > 0$:

$$\frac{a - b}{\sqrt{a} - \sqrt{b}}$$

3. Effort & Scaffolding | Level 3: Advanced Binomials (9 min)

[Level 3: Hard – Cooperative Groups]

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Task 2: Algebraic Cancellation

Simplify fully for $a \neq b, a, b > 0$:

$$\frac{a - b}{\sqrt{a} - \sqrt{b}}$$

Solutions

Task 1: Rationalizing yields $2\sqrt{5}$.

Task 2: Rationalizing yields $\sqrt{a} + \sqrt{b}$.

3. Effort & Scaffolding | Level 4: Master Challenge (9 min)

[Level 4: Expert Challenge – Trinomial Denominators]

Goal: Apply multi-step rationalization to denominators with three terms.

Rationalize: $\frac{1}{\sqrt{2} + \sqrt{3} + \sqrt{5}}$

3. Effort & Scaffolding | Level 4: Master Challenge (9 min)

[Level 4: Expert Challenge – Trinomial Denominators]

Goal: Apply multi-step rationalization to denominators with three terms.

Rationalize: $\frac{1}{\sqrt{2} + \sqrt{3} + \sqrt{5}}$

Solution

Rationalizing yields: $\frac{2\sqrt{3}+3\sqrt{2}-\sqrt{30}}{12}$.

4. Recognition & Feedback | Peer Assessment Protocol

Notebook Exchange Protocol – Level 3 (7 min):

Layer 1 (Praise): Highlight clear step-by-step layout and correct conjugate choice.

Layer 2 (Critique): Identify missing simplifications, sign errors, or unreduced fractions.

Layer 3 (Validation): Sign and return the notebook to validate peer success.

Metacognitive Exit Ticket

Submit your peer-signed work and write and reflect on your rationalization strategy.

Radical Form	Rationalizing Multiplier	Resulting Denominator
Square Root: $\frac{a}{c\sqrt{b}}$	$\frac{\sqrt{b}}{\sqrt{b}}$	$c \cdot b$
n -th Root: $\frac{a}{c\sqrt[n]{b^k}}$	$\frac{\sqrt[n]{b^{n-k}}}{\sqrt[n]{b^{n-k}}}$	$c \cdot b$
Binomial: $\frac{a}{\sqrt{b} \pm \sqrt{c}}$	Conjugate $\frac{\sqrt{b} \mp \sqrt{c}}{\sqrt{b} \mp \sqrt{c}}$	$b - c$

Congratulations on mastering advanced rationalization!